Homework #3 ~ Fawzia Jassim

#1 First we would ignore all the spaces so it would be WRITETHELETTER and the key stream is: 5 17 9 15 13 8 21 3 11 6 18 14 10 25

Part a) W=22, R=17, I=8, T=19, E=4, T=19, H=7, E=4, L=11, E=4, T=19, T=19, E=4, R=17

1. W(22) + 5 = 27 → 27 mod 26 = 1 → B
2. R(17) + 17 = 34 → 8 → I
3. I(8) + 9 = 17 → R
4. T(19) + 15 = 34 → 8 → I
5. E(4) + 13 = 17 → R
6. T(19) + 8 = 27 → 1 → B
7. H(7) + 21 = 28 → 2 → C
8. E(4) + 3 = 7 → H
9. L(11) + 11 = 22 → W
10. E(4) + 6 = 10 → K
11. T(19) + 18 = 37 → 11 → L
12. T(19) + 14 = 33 → 7 → H
13. E(4) + 10 = 14 → O
14. R(17) + 25 = 42 → 16 → Q

BIRIRBCHWKLHOQ

Part b) STANDONBALCONY → the decryption formula is → P = (C - K) mod 26 and to find the needed key stream we would rearrange it to K = (C - P) mod 26

S=18, T=19, A=0, N=13, D=3, O=14, N=13, B=1, A=0, L=11, C=2, O=14, N=13, Y=24

CS Scanned with CamScanner

1. $1-18=-17 \to 9$
2. $8-19=-11 \to 15$
3. $17-0=17$
4. $8-13=-5 \to 21$
5. $17-3=14$
6. $1-14=-13 \to 13$
7. $2-13=-11 \to 15$
8. $7-1=6$
9. $22-0=22$
10. $10-11=-1 \to 25$
11. $11-2=9$
12. $7-14=-7 \to 19$
13. $14-3=1$
14. $16-24=-8 \to 18$

So, the decryption key stream is: 9 15 17 21 14 13 15 6 22 25 9 19 1 18

#2. $p=262643$, $g=9563$, private key $K_{priv}=2824$

a) First, in El Gamal, I know that the public key has three parts $\to (p, g, y)$ where $y = g^{K_{priv}} \bmod p$.
So, first I take the formula $y = 9563^{2824} \bmod 262643$
Now the exponent 2824 is very large, so I can't really multiply 9563 by itself 2824 times. But instead I use modular exponentiation. That means I keep squaring the number and reducing mod 262643 each time so the number doesn't get too big.

Scanned with CamScanner

So I compute powers of 9563 -> 9563^1 mod 2626643 = 9563 then I square it -> 9563^2 mod 2626643 then I keep squaring 9563^4, 9563^8, 9563^{16}, 9563^{32}... Each time I reduce mod 2626643 so the number stays small. Then I write 2824 in binary form so I know which powers I need: 2824 = 2048 + 512 + 256 + 8 so that means $9563^{2824} = 9563^{2048} \cdot 9563^{512} \cdot 9563^{256} \cdot 9563^{8}$ after multiplying those powers together and reducing mod 2626643 at each step to get -> y = 137186. So, now I can write the public key and the public key is -> (2626643, 9563, 137186) and the private key stays -> 2824

b) Okay so now K must be between 1 and p-2 and be relatively prime to p-1. So now p-1 = 2626643 - 1 = 2626642. Now I check gcd -> since 2626642 is even and 2 is even, they share a factor of 2. So -> gcd (2, 2626642) ≠1 which means 2 is not relatively prime to 2626642. So K = 2 can't be used.
Now I would check K = 5 -> 2626642 is not divisible by 5 (it doesn't end in 0 or 5)
So gcd (5, 2626642) = 1 that means 5 is relatively prime to 2626642. So K=5 can be used. So basically K=2 -> is not allowed and K=5 -> allowed.

c) (c_1, c_2) where $c_1 = g^k$ mod p and $c_2 = m \cdot y^k$ mod p
So now K=15, g=9563, y=137186, p=2626643
$c_1 = 9563^{15}$ mod 2626643 so now I break 15 into binary -> 15 = 8+4+2+1 so $9563^{15} = 9563^8 \cdot 9563^4 \cdot 9563^2 \cdot 9563^1$ then again I repeated squaring and

CS Scanned with CamScanner

reduce mod 262643 each time then after multiplying all needed powers and reducing $\rightarrow c_1 = 183847$

So, now 137186^{15} mod 262643 then I again break 15 into $\rightarrow 8+4+2+1$ then multiply those powers together and reduce mod 262643 each time and after that I would get $\rightarrow y^{15}$ mod $p = 104522$

Encrypt $m_1 = 1008 \rightarrow c_2 = 1008 \times 104522$ mod 262643

I first multiply $1008 \times 104522 = 105358176 \rightarrow$ now I would reduce mod 262643 $\rightarrow 105358176$ mod $262643 = 231021$ so basically the ciphertext is $(183847, 231021)$

Now encrypt $m_2 = 2111 \rightarrow c_2 = 2111 \times 104522$ mod 262643 so first I multiply $\rightarrow 2111 \times 104522 \rightarrow$ $2111 = 2000 + 100 + 10 + 1$ then multiply 104522×2000 $\rightarrow 104522 \times 2 = 209044 = 209044000$

$104522 \times 100 = 10452200$

$104522 \times 10 = 1045220$

$104522 \times 1 = 104522$

So now $209044000 + 10452200 + 1045220 + 104522$

So first $209044000 + 10452200 = 219496200$

then $219496200 + 1045220 = 220541420$

$\hookrightarrow 220541420 + 104522 = 220645942$

So finally $\rightarrow 2111 \times 104522 = 220645942$

#3. $(p, g, y) = (23, 3, 12)$

First $y = g^x$ mod p where x is the private key.

So, I need to find x such that: $3^x = 12$ (mod 23)

Basically what I do is I start testing powers of

CS Scanned with CamScanner

3 mod 23 until I get 12.

$3^1 = 3 \pmod{23} \rightarrow 3^2 = 9 \pmod{23} \rightarrow 3^3 = 27 = 4 \pmod{23}$ because $27 - 23 = 4 \rightarrow 3^4 = 3^3 \cdot 3 = 4 \cdot 3 = 12 \pmod{23}$ now I got 12. So private key is $x = 4$.

b) (c_1, c_2) where $c_1 = g^k \bmod p$ and $c_2 = m \cdot y^k \bmod p$

$k = 5,\ g = 3,\ y = 12,\ p = 23,\ m = 45$

Since El Gamal works mod p, I first reduce 45 mod 23.

$45 \div 23 = 1$ remainder 22 so $45 \equiv 22 \pmod{23}$

so I will encrypt 22 instead.

$c_1 = 3^5 \bmod 23 \rightarrow 3^4 = 12$ so $3^5 = 3^4 \cdot 3 = 12 \cdot 3 = 36$ now reduce mod 23 $\rightarrow 36 - 23 = 13$ so $c_1 = 13$

$y^k = 12^5 \bmod 23 \rightarrow 12^2 = 144$ then reduce mod 23 $\rightarrow$ $144 \div 23 = 6$ remainder 6 so $12^2 \equiv 6$ now $12^4 = (12^2)^2 = 6^2 = 36$ reduce mod 23 $\rightarrow 36 - 23 = 13$ so $12^4 \equiv 13$ now $12^5 = 12^4 \cdot 12 = 13 \cdot 12 = 156$ reduce mod 23 $\rightarrow 156 \div 23 =$ 6 remainder 18 because $23 \times 6 = 138$, and $156 - 138 = 18$ so $12^5 \equiv 18$

$c_2 = m \cdot y^k \bmod p \rightarrow$ we already reduced m to 22 and I found $\rightarrow y^k \equiv 18$ so $c_2 = 22 \cdot 18 \rightarrow 22 \cdot 18 = 396$ now reduce mod 23 $\rightarrow 396 \div 23 = 17$ remainder 5 because $23 \times 17 = 391$ and $396 - 391 = 5$ so $c_2 = 5$

So the private key is $\rightarrow x = 4$ and encrypted message is $(c_1, c_2) = (13, 5)$

#4. In a simple substitution cipher using a sentence as the key, I first write all the letters from the key sentence but I would remove spaces, remove punctuation, remove any repeated letters, and keep

Scanned with CamScanner

only the first time each letter appears, so I will be
THEQUICKBROWNFOXJUMPSOVERTHELAZYDOG
Now I remove any duplicates and after checking this already contains all 26 letters of the alphabet exactly once (because this sentence is a pangram) so this becomes my cipher alphabet.
Now I write them one under the order!

ABCDEFGHIJKLOMNOPQRSTUVWXYZ
THEQUICKBROWNFXJMPSVLAZYDG

So A->T, B->H, C->E, D->Q, E->U, F->I, G->C, H->K, I->B, J->R, K->O, L->W, M->N, N->F, O->X, P->J, Q->M, R->P, S->S, T->V, U->L, V->A, W->Z, X->Y, Y->D, Z->G

MAY THE FORCE B WITH YOU
so M->N, A->T, Y->D, T->V, H->K, E->U, F->I, O->X, R->P, C->E, E->U, B->H, E->U, W->Z, I->B, T->V, H->K, H->K, Y->D, O->X, U->L
So ->NTDVKUIXPEUHUZBVKDXL

#5. First, in a colomnar transposition cipher, the key tells us that the order of the colomns. so the key is: BROAD -> in alphabetical order it would be ABDOR
B (position 1), R (position 2), O (position 3), A (position 4), D (position 5) and in alphabetical order -> A -> column 4, B -> column 1, D -> column 5, O -> column 3, R -> column 2
So 4, 1, 5, 3, 2

CS Scanned with CamScanner

EAULSTIUL FEFIBI is 15 letters ....
Since the key BROAD has 5 letters, that means there are 5 columns so 15 ÷ 5 = 3 rows so the table must be 3 rows × 5 columns.
Since in encryption, the columns were read in alphabetical key order 4, 1, 5, 3, 2 so now I reverse that process. Since there are 3 rows, each column must contain 3 letters. now I divide the ciphertext into 5 groups of 3 letters so basically:

EAU
LST
IUL
FEF
IBI

So now I place them back into the correct columns:

Column 4 → EAU
Column 1 → LST
Column 5 → IUL
Column 3 → FEF
Column 2 → IBI

columns:
1 = LST
2 = IBI
3 = FEF
4 = EAU
5 = IUL

Row 1:	Row 2:	Row 3:
Column 1 → L	Column 1 → S	Column 1 → T
Column 2 → I	Column 2 → B	Column 2 → I
Column 3 → F	Column 3 → E	Column 3 → F
Column 4 → E	Column 4 → A	Column 4 → U
Column 5 → I	Column 5 → U	Column 5 → L
LIFEI	SBEAU	TIFUL

CS Scanned with CamScanner

LIFEIS BEAUTIFUL

So, I first arranged the key BOARD in alphabetical order to determine the column reading order. Then I counted the ciphertext letters and found there are 15 letters, so with 5 columns there are must be 3 rows. Since the columns were read in the order 4, 1, 5, 3, 2 during encryption, I split the ciphertext into five groups of 3 letters and placed them back into the correct columns. Finally, I got LIFEISBEAUTIFUL.

#6. OH HAD I THE WINGS OF A DOVE →
AVBSHUHBWAUBAKSROXGZQ → both are 20 letters
$C = (P+K) \bmod 26$ → to find the key it would be
$K = (C-P) \bmod 26$ so for each letter, I would subtract the plaintext letter number from the ciphertext letter number
So O=14, H=7, H=7, A=0, D=3, I=8, T=19, H=7, E=4, W=22, I=8, N=13, G=6, S=18, O=14, F=5, A=0, D=3, O=14, V=21, E=4 → this is the plaintext to numbers
Now the ciphertext to numbers → A=0, V=21, B=1, S=18, H=7, U=20, H=7, B=1, W=22, A=0, U=20, B=1, A=0, K=10, S=18, R=17, O=14, X=23, G=6, Z=25, Q=16
Now I do $K = (C-P) \bmod 26$ → 1. 0 - 14 = -14 → 12

2. 21 - 7 = 14
3. 1 - 7 = -6 → 20
4. 18 - 0 = 18
5. 7 - 3 = 4
6. 20 - 8 = 12
7. 1 - 7 = -6 → 20
8. 1 - 7 = -6 → 20
9. 22 - 4 = 18
10. 0 - 22 = -22 → 4
11. 20 - 8 = 12
12. 1 - 13 = -12 → 14
13. 0 - 6 = -6 → 20

CS Scanned with CamScanner

14. 10 - 18 = -8 -> 18
15. 18 - 14 = 4
16. 17 - 5 = 12
17. 14 - 0 = 14
18. 23 - 3 = 20
19. 6 - 14 = -8 -> 18
20. 25 - 21 = 4
21. 16 - 4 = 12

Now to convert those numbers back to letters ->
12 = M, 14 = O, 20 = U, 18 = S, 4 = E, 12 = M, 14 = O, 20 = U, 18 = S, 4 = E, 12 = M, 14 = O, 20 = U, 18 = S, 4 = E, 12 = M, 14 = O, 20 = U, 18 = S, 4 = E, 12 = M

MOUSEMOUSEMOUSEMOUSEM -> repeating pattern of the word Mouse

↳ So to find the key, I used the Vingenère formula K = (C - P) mod 26. I converted both plaintext and ciphertext letters into numbers (A = 0 to Z = 25), then I subtracted each plaintext value from the corresponding ciphertext value and reduced mod 26. After converting the results back to letters, I observed the repeating pattern which is the key used in the encryption.

#7. In this program, the login system builds an SQL query using whatever the user types into the id, password, and pin fields. The program takes that input and directly places it inside the SQL query without checking it first. Because of this, someone can type special SQL commands instead of normal input. The important line in the program builds the query like this -> SELECT accounts FROM users WHERE id = '[id]' AND pass = '[pwd]' AND pin = [pin] and then the attackers enters this in the id field which is -> 'or 5 = 5--

a) I first substitute the attacker's input into the query
↳ SELECT accounts FROM users WHERE id = '' or 5 = 5 -- ' AND pass = '[pwd]' AND pin = [pin] -> first the ' closes the id field then or 5 = 5 is added as a

CS Scanned with CamScanner

now condition. Then -- Starts a comment in SQL. Everything after -- is ignored by the database. So the database effectively runs this query -> SELECT accounts FROM users WHERE id = '' OR 5=5

b) id = '' OR 5=5 -> the part 5=5 is always true. Because one part of the condition is always true, the whole WHERE condition becomes true. This means the database will return records even if the id, password, and pin are wrong. Since the database returns results, the program will run: displayAccounts(results); So the attacker successfully logs in without knowing the correct password or pin. The login security is completely bypassed.

#8. The application runs this query to show employee names from a department ->

SELECT name
FROM employee
WHERE department = ' '

The program takes whatever the user types for the department and places it between the quotes. Then now the attacker enters this -> finance' UNION SELECT date_of_birth, address FROM employee --

So first I insert the attacker's input into the query is SELECT name
FROM employee
WHERE department = 'finance' UNION SELECT date_of_birth, address FROM employee --'

So, here is what happens -> so first the ' after finance closes the department value. Then the attacker adds UNION SELECT date_of_birth, address FROM employee.

Scanned with CamScanner

Finally, -- comments out the rest of the original query.
So, the database basically effectively runs like this:

```
SELECT name
FROM employee
WHERE department = 'finance'
UNION
SELECT date_of_birth, address FROM employee
```

The UNION operator combines the results of two SELECT statements. So, instead of only returning employee names from the finance department, the database also returns -> date_of_birth and address from the employee table.

So basically, originally the user was only allowed to see employee names. But because of this SQL injection attack, the query now returns important information like employee's dates of birth and addresses.

So the attacker is able to see private data that they should not have access to. This is an example of a UNION SQL injection attack, where attackers use UNION to pull extra data from the database.

CS Scanned with CamScanner